

Python Project Roadmap (Based on Day 1-8 Learning)

Goal: Combine all learned Python concepts through mini projects and build portfolio-worthy work.

Project 1: BudgetWise (7 days)

Skills: Variables, Loops, Functions, Dictionaries, File Handling, Exception Handling

Tasks: Menu system -> Add expense -> Store list/dict -> Save to file -> Read file -> Budget warning

Project 2: Student Marks Analyzer (3-4 days)

Skills: Lists, Dictionaries, NumPy, Conditions

Tasks: Input marks -> NumPy array -> Average -> Highest/Lowest -> Grade assignment

Project 3: Library Management System (5-6 days)

Skills: Classes, Objects, Inheritance, Exception Handling

Tasks: Create Book class -> Add books -> Borrow logic -> Return logic -> Raise exceptions

Project 4: Study Streak Tracker (4-5 days)

Skills: Lists, File Handling, Functions, Loops

Tasks: Input study hours -> Store file -> Total hours -> Average -> Streak count

Project 5: Employee Salary Analyzer (NumPy)

Skills: NumPy arrays, Slicing, Matrix Ops, Indexing

Tasks: Create employee matrix -> Slice salary column -> Max/Min -> Average -> Employee with highest salary

Recommended Order:

Week 1: BudgetWise | Week 2: Student Marks Analyzer | Week 3: Library System | Week 4: Salary Analyzer

Rule: 80% Projects + 20% Tutorials